Pass-Through, Welfare, and Incidence under Product Returns

Draft notes

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Abstract

With product returns, local welfare and incidence weight kept units rather than gross sales, and pass-through is unchanged once list price is replaced by effective marginal revenue.

1 Setting

A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u-p,\, -\kappa_i)\big].$$

For purchase, all consumer-side heterogeneity summarizes into willingness to pay v_i, defined by $s_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, $F(v) = \Pr(v_i \leq v)$, gross demand is $D(p) = 1 - F(p)$, and share $r(p)$ of gross units sold at p are returned.

The product market is perfectly competitive, producer j takes p as given, chooses output at cost $C_j(q_j)$, and pays handling cost $h > 0$ on each returned unit.

2 Consumer side

Write $s(v,p)$ for expected surplus at list price p among types with willingness to pay v, so $s(v,v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty s(v,p)\, dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -s(p,p)\, f(p) + \int_p^\infty s_p(v,p)\, dF(v),$$

where $s_p(v,p) \equiv \partial s(v,p)/\partial p$ and f is the density of v_i when F is differentiable. On the extensive margin, $s(p,p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v,p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$s_p(v,p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty s_p(v,p)\,dF(v) = -D(p)\big(1-r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1-r(p)\big)\,dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1-r(p)\big)\,dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D,r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1-r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p})=0$ and $CS(\bar{p})=0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1-r(z)\big)\,dz. \tag{3}$$

3 Supply side

Returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p,h) \equiv p - r(p)(p+h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p,h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \ge 0}\ \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \ \text{ (if } q_j > 0\text{)},$$

with $C_j(0)=0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\,dz. \tag{6}$$

At equilibrium $\mu = \mu(p,h)$, $PS = \int_0^{\mu(p,h)} S(z)\,dz$.

4 Failure of tax neutrality

Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. With returns, we ask whether that equivalence survives by writing the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p+h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p-\tau) - r(p-\tau)(p-\tau+h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p-\tau$ yields

$$r(p)(p+h) = r(p-\tau)(p-\tau+h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.*[1]

Tax neutrality breaks down with product returns because the refunded amount—and therefore the keep-or-return decision—depends on which side of the market remits the tax. Under seller remittance, the checkout price (which includes the tax) is refunded, so consumers decide whether to return based on that tax-inclusive price. Under buyer remittance, the seller posts a pre-tax list price, only that list price is refunded, and the return decision is based on this lower, tax-exclusive price. Even though the total checkout price and the tax are identical in both regimes, the return rate and the seller's refund liability per return are evaluated at different prices (p versus $p - \tau$).

The direction in which the return rate changes is ambiguous—a higher price can make each individual more likely to return but can also select consumers with stronger purchase intentions who are less likely to return—so $r(p)$ may be above or below $r(p-\tau)$. Regardless of the direction, the seller's effective marginal revenue generally differs, so the two tax regimes produce distinct market-clearing conditions and are not equivalent. The fundamental cause is that the purchase decision is always based on the checkout price, while the return decision uses the tax-inclusive price under seller remittance and the tax-exclusive price under buyer remittance, creating a wedge that alters equilibrium.

[1]Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p-\tau$. That holds for all taxes only if $r(p)(p+h)$ is constant in posted list price, which requires $r(p) = C/(p+h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

5 Pass-through

Write $Q \equiv D(p)$. We first study pass-through for two cost shocks: a uniform increase in marginal production cost (denoted c) and an increase in the per-unit handling cost h; the same framework is then extended to a per-unit tax. A uniform production-cost increase means adding one dollar to each firm's marginal cost $C_j'(q_j)$ at every output level. Production cost and handling cost hit equilibrium on different margins. Production cost shifts market supply at a given μ without moving μ at fixed list price. Handling cost moves μ through (4) at fixed list price without shifting supply at a given μ.

Equilibrium satisfies

$$m(p,x) \equiv D(p) - S(\mu(p,x),x) = 0, \qquad x \in \{c,h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

where $m_p \equiv \partial m/\partial p$ and $m_x \equiv \partial m/\partial x$. Since $m = D - S$,

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

with $\mu_p \equiv d\mu/dp$, $\mu_x \equiv \partial\mu/\partial x$ at fixed p, $S'(\mu) \equiv dS/d\mu$, and $S_x \equiv \partial S/\partial x$ at fixed μ. Hence

$$\rho_x = \frac{S'(\mu)\,\mu_x + S_x}{D'(p) - S'(\mu)\,\mu_p}. \tag{13}$$

At fixed list price p, a one-dollar parallel rise in production cost shifts supply at fixed μ, so $\mu_c = 0$ and $S_c = -S'(\mu)$. A one-dollar rise in handling cost moves μ by $r(p)$ per gross unit sold and does not shift supply at fixed μ, so $\mu_h = -r(p)$ and $S_h = 0$. Hence

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

Substituting into (13) yields

Result 2 (Pass-through of production and handling costs)**.**

$$\frac{\rho_h}{\rho_c} = r(p), \tag{15}$$

with pass-through into list price

$$\rho_c = \frac{-S'(\mu)}{D'(p) - S'(\mu)\,\mu_p}, \qquad \rho_h = r(p)\,\rho_c. \tag{16}$$

A one-dollar increase in production cost is equivalent to decreasing μ by one unit. Market clearing $D(p) = S(\mu)$ translates a given change in μ into list price through the same local comparative statics. Handling moves μ by only $r(p)$ per dollar relative to that unit shift, so $\rho_h/\rho_c = r(p)$. When $r \equiv 0$, handling does not move μ and ρ_c coincides with the textbook competitive pass-through formula.

Write p^s for checkout price under seller remittance and p^b for checkout price under buyer remittance; let $\rho_\tau^s \equiv dp^s/d\tau$ and $\rho_\tau^b \equiv dp^b/d\tau$. For a local change at $\tau = 0$, tax pass-through is (13) with $S_\tau = 0$: the denominator uses the same μ_p as above, and the only difference across remittance regimes is $\mu^s_\tau \equiv \partial\mu^s/\partial\tau$ versus $\mu^b_\tau \equiv \partial\mu^b/\partial\tau$ at fixed checkout price:

$$\rho^s_\tau = \frac{S'(\mu)\,\mu^s_\tau}{D'(p^s) - S'(\mu)\,\mu_p}, \qquad \rho^b_\tau = \frac{S'(\mu)\,\mu^b_\tau}{D'(p^b) - S'(\mu)\,\mu_p}. \tag{17}$$

At $\tau = 0$,

$$\mu^s_\tau = -1, \qquad \mu^b_\tau = r(p^b) + (p^b + h)\,r'(p^b) - 1. \tag{18}$$

Under seller remittance, $\mu^s_\tau = -1$ at fixed checkout price, so $\rho^s_\tau = \rho_c$. Under buyer remittance, μ^b_τ differs, so tax pass-through generally differs from ρ_c.

6 Incidence

Consumer surplus responds to a cost shock only through pass-through,

$$\frac{dCS}{dx} = -D(p)\big(1 - r(p)\big)\,\rho_x. \tag{19}$$

Producer surplus is $PS = \int_0^\mu S(\tilde\mu)\,d\tilde\mu$ at equilibrium. Differentiating along the shock gives

$$\frac{dPS}{dx} = \int_0^\mu S_x(\tilde\mu, x)\,d\tilde\mu + Q\big(\mu_p\,\rho_x + \mu_x\big), \tag{20}$$

using $PS'(\mu) = Q$. For production cost, $\mu_c = 0$ and $S_c = -S'(\tilde\mu)$, so

$$\int_0^\mu S_c(\tilde\mu)\,d\tilde\mu = -Q, \qquad \frac{dPS}{dc} = Q\big(\mu_p\,\rho_c - 1\big). \tag{21}$$

For handling, $S_h = 0$, so

$$\frac{dPS}{dh} = Q\big(\mu_p\,\rho_h + \mu_h\big). \tag{22}$$

The supply-shift term equals the drop in industry quantity at fixed μ, the same $-Q$ that would appear if μ fell by one dollar at fixed p. Both shocks therefore satisfy

$$\frac{dPS}{dx} = Q\big(\mu_p\,\rho_x + \mu_x\big), \tag{23}$$

with $\mu_x = -1$ for production cost and $\mu_x = \mu_h$ for handling.

Write $I_x \equiv |dCS/dx|/|dPS/dx|$. For both shocks,

$$I_c = I_h = \frac{(1 - r(p))\,S'(\mu)}{-D'(p)}. \tag{24}$$